Yashraj Wani

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SUMMARY

Computational biophysicist developing physics-based simulation and machine learning methods to understand how the material properties of biomolecular condensates govern molecular transport and phase behavior. Postdoctoral researcher at Princeton University (CZI Exploratory Cell Networks Project; Gordon and Betty Moore Foundation), working on non-equilibrium frameworks for flux through condensates and ESM2-based models that predict phase diagrams for intrinsically disordered proteins at proteome scale – reducing prediction time from ~5 CPU-days per protein to seconds. PhD in Physics, Magna cum Laude, from Johannes Gutenberg University Mainz; contributor to the open-source GPU-accelerated simulation package HOOMD-blue; 4 first- or co-first-author publications plus 1 under review in peer-reviewed journals.

TECHNICAL SKILLS

AI & Deep Learning: PyTorch, TensorFlow, protein language models (ESM2), high-throughput model inference

Programming: Python, C++, Bash, R, Git

Simulation & Modeling: LAMMPS, HOOMD-blue (contributor), VMD, Ovito, molecular dynamics, coarse-grained and all-atom force fields

HPC & Compute: Slurm, large-scale GPU cluster computing, large-scale trajectory analysis

Scientific Communication: 4 published + 1 under review first/co-first-author papers, international conference presentations, cross-disciplinary collaboration

EXPERIENCE

Postdoctoral Researcher – Princeton University, USA *Oct 2024 - Present*

Funded by the CZI (Exploratory Cell Networks Project) and the Gordon and Betty Moore foundation.

- Leveraged ESM2 embeddings to train a phase diagram prediction model on ~2,000 intrinsically disordered proteins from the human IDRome; validated on held-out test set (20%) with mean absolute error of ~7K in critical temperature (within noise of simulation ground truth) – reducing prediction time from ~5 CPU-days per protein to seconds and enabling proteome-scale screening. Web tool for single-sequence and batch prediction in development; manuscript in preparation.
- Developed a non-equilibrium framework for directed molecular flux through biomolecular condensates: pore size distribution governs transport rates, and high-affinity interactions assist flux through a handoff mechanism. Manuscript under review (Wani et al., bioRxiv 2026).
- Using MD simulations to map how tetrapeptide chemistry (four residues spanning attractive, spacer, positive, and negative interactions) governs condensate partitioning and stability; charge and hydrophobicity are the primary determinants.

Doctoral Researcher – Johannes Gutenberg University of Mainz, Germany *Nov 2020 - Sept 2024*

Funded by Deutsche Forschungsgemeinschaft (German Research Foundation)

- Designed and implemented a new module in HOOMD-blue (GPU-accelerated molecular simulation package with 1,000+ citations) extending its infrastructure for droplet simulation geometries – open-source contribution available to the broader community.
- Built a mesoscale computational framework for colloidal suspensions with hydrodynamic interactions via multiparticle collision dynamics, supporting shape-anisotropic nanoparticles; generated and analyzed large-scale simulation trajectories.
- Led investigation into nanoparticle self-assembly in evaporating droplets, validated against experimental data; resulted in 4 first- or co-first-author and 2 co-author publications.

Research Intern – Bhabha Atomic Research Center, Mumbai, India *May - July 2018*

- Studied density functional theory (DFT) and electronic structure calculations; gained exposure to quantum chemical methods.

EDUCATION

Ph.D., Physics – Johannes Gutenberg University of Mainz, Germany *2020 - 2024*

Magna cum Laude (GPA: 4.0/4.0, US equivalent)

BS-MS Dual Degree, Physics – IISER Bhopal, India *2015 - 2020*

GPA: 3.5/4.0 (US equivalent)

PUBLICATIONS

Lead / Co-First Author

- Wani et al. Non-equilibrium modeling of directed flux through biomolecular condensates. Under review, bioRxiv (2026). doi:10.64898/2026.02.03.703649
- Wani et al. Mesoscale simulations of diffusion and sedimentation in shape-anisotropic nanoparticle suspensions. Soft Matter, 20.19 (2024).
- Wani et al. Coarse-Grained Simulations of Columnar Ionic Liquid Crystals: Comparison with Experiments. J. Phys. Chem. B, 128.34 (2024).
- Wani et al. Structure formation in supraparticles composed of spherical and elongated particles. Langmuir, 40.1 (2023).
- Wani et al. Diffusion and sedimentation in colloidal suspensions using multiparticle collision dynamics with a discrete particle model. J. Chem. Phys., 156.2 (2022).

Co-Author

- Tuning the morphology and structure of supraparticles composed of ellipsoids. Langmuir, 41.11 (2025).
- Exploring the role of hydrodynamic interactions in spherically confined drying colloidal suspensions. J. Chem. Phys., 162.15 (2025).

SELECTED TALKS & CONFERENCES

PR Condensate Day – Physical determinants of flux through condensates, oral, Sept 2026
Biophysical Society Annual Meeting – poster, Feb 2024
Auburn University Seminar – Watching Droplets Dry, invited oral, June 2023
97th ACS Colloid and Surface Science Symposium – oral, June 2023
Deutschen Physikalischen Gesellschaft (Condensed Matter) – oral, March 2023
36th European Colloid and Interface Society – poster, Sept 2022
APS March Meeting – poster, March 2022

REFERENCES

Prof. Jerelle Joseph – Dept. of Chemical & Biological Engineering, Princeton University
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